

# Craft, Kinetics & Systems Code.

Engineering a living, interactive design portfolio from scratch: pairing Swiss editorial typography with real-time browser canvas physics, 120fps motion engineering, and an automated ATS resume pipeline.

DESIGNER & ENGINEER

Ritvik Chunamari

ARTIFACT TYPE

Interactive Web Portfolio

TECH STACK

React 19 • Vite • Canvas • Framer Motion

LIVE DEPLOYMENT

[ritvikchunamari.github.io](https://ritvikchunamari.github.io)

# Why Rebuild? Beyond the Cookie-Cutter Template

Design portfolios often suffer from two extremes: lifeless static PDF grids or bloated, laggy no-code templates. My goal was to create a digital home where the medium is an active proof of craft.

## PILLAR 01

### Code as Design Material

Moving away from black-box website builders. Writing the core architecture in clean React and TypeScript allowed tactile control over animation curves, scroll performance, and fluid geometry.

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◆ Full Architectural Autonomy

## PILLAR 02

### Tactile Kinetics & Physics

Static screens don't capture how digital products feel. The portfolio features an interactive antigravity particle canvas that responds to cursor velocity, inertia, and user proximity fields.

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◆ Real-Time Canvas Particle Engine

## PILLAR 03

### Live, Usable Artifact

Built as a real artifact that recruiters, peers, and clients can try, stress-test, and interact with across devices. Includes live deep-dive case studies and an automated 1-click vector PDF resume.

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◆ 100% Responsive & Publicly Deployed

# Anatomy of the Artifact: Four Core Systems

A modular design system structured around four interconnected components engineered for maximum responsiveness and high-frequency tactile interaction.

## SYSTEM 01

### Antigravity Canvas

Interactive background simulation featuring autonomous floating particles and reactive velocity-based cursor repulsion.

HTML5 2D Canvas • 60 FPS

## SYSTEM 02

### Swiss Hero Layout

Ultra-bold typographic display calibrated to fill the full margin allocation across mobile, tablet, and widescreen monitors.

Dynamic Clamp() Typography

## SYSTEM 03

### Case Study Deep-Dives

Detailed case studies for enterprise UX: USPS Fleet, Nexus AI node editor, SOC Security Agency, and Mumbai 2028 branding.

4 Production Case Studies

## SYSTEM 04

### Automated ATS Resume

Headless Chromium print pipeline generating an exact 2-page, 100% vector PDF resume matching the site's typography tokens.

Taleo & Lever Parseable

# Where It Broke: The Reality of Browser Constraints

Designing in Figma is effortless; compiling into a high-performance browser runtime exposed critical edge-cases in animation budgeting, touch friction, and viewport geometry.

## × BOTTLENECK 01 • SEVERE SCROLL STUTTER

### 3D RotateX & Continuous State Triggers

Attempted 3D perspective tilts on scroll cards. Continuous React state recalculations during scroll flooded the event loop, dropping frames to <30fps on high-DPI displays and causing visible stutter.

## × BOTTLENECK 02 • TOUCH DEVICE CURSOR CLUTTER

### Desktop Physics on Touchscreens

The kinetic cursor physics that felt snappy with a mouse became intrusive on mobile and tablet touchscreens, conflicting with native swipe inertia and eating up mobile battery/GPU cycles.

## × BOTTLENECK 03 • MOBILE TYPOGRAPHY OVERFLOW

### Name Breaking & Awkward Wrapping

The name headline "CHUNAMARI" broke across lines or overflowed horizontally on narrower mobile screens (375px–390px), failing the editorial rule of keeping "Ritvik" on line 1 and "Chunamari" on line 2.

## × BOTTLENECK 04 • THE 3-PAGE RESUME SPILL

### Awkward 6-Line Orphan Page

Initial resume PDF export bled into an awkward 3rd page containing only 6 lines of personal insights, breaking recruiter reading ergonomics and failing standard ATS parse budgets.

# Engineering the Fixes: Optimization as Craft

Fixing these bottlenecks required treating performance not as a late cleanup task, but as a primary design constraint that directly sharpened the visual language.

## FIX 01 • PERFORMANCE

### Zero-Render DOM Scroll

Eliminated 3D perspective transforms and replaced React state scroll hooks with direct CSS custom property updates managed inside a single `requestAnimationFrame` loop. Achieved a locked 120fps scroll experience.

◆ 120 FPS Locked • Zero Layout Thrashing

## FIX 02 • TOUCH UX

### Touch-Aware Soft Fade

Implemented pointer media queries like `@media (hover: none)` and dynamic touch listeners. The cursor and particle physics smoothly fade to 0 opacity the moment a touch gesture starts, restoring native mobile scrolling.

◆ Frictionless Mobile Ergonomics

## FIX 03 • TYPOGRAPHY & PDF

### Fluid Math & 2-Page Budget

Calibrated fluid `clamp()` typography formulas ensuring "Ritvik" and "Chunamari" occupy 100% of the allocated mobile width with zero overflow. Rebalanced resume print geometry into an exact 2-page vector PDF.

◆ Perfect Edge-to-Edge Typography

# What I Learned: Beyond the Mockup

Three fundamental truths about digital product design that only revealed themselves through hands-on code and production deployment.

## TAKEAWAY 01

### Performance IS User Experience

A beautiful design that stutters or drops frames immediately destroys user trust. Fluidity, input responsiveness, and latency communicate high craftsmanship far more deeply than decorative graphics ever could.

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◆ 60/120fps is a Design Requirement

## TAKEAWAY 02

### Restraint Over Spectacle

The initial urge when building an interactive portfolio is to cram in every animation, particle, and 3D angle. True design discipline lies in removing decorative noise so the case studies and typography take center stage.

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◆ Subtlety Elevates Content

## TAKEAWAY 03

### The Power of the Hybrid Mindset

Designing with code creates an intuition for browser execution—reflows, repaint cycles, and memory management. When you understand how the browser thinks, your UI designs become inherently cleaner and more feasible.

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◆ Designer-Engineer Symbiosis

# Studio Share: Where I Want the Room's Help

Live artifact available at [ritvikchunamari.github.io](https://ritvikchunamari.github.io). As you test the site on your laptops and phones, I'd love feedback on these three specific areas:

## QUESTION 01 • INTERACTION PACING

### Tactile Delight vs. Content Focus

Does the interactive antigravity cursor feel like a rewarding, playful detail, or does it risk competing with readability and case study navigation?

👉 Looking for: Is the physics tension balanced or overly noisy?

## QUESTION 02 • CROSS-PLATFORM FEEL

### Desktop Kinetics vs. Mobile Typography

On desktop, the site's brand is driven by mouse interaction; on mobile, it transitions into pure bold typography. Does the site's identity still feel cohesive across both devices?

👉 Looking for: Does mobile retain the same high-craft personality?

## QUESTION 03 • CASE STUDY SCANNABILITY

### Visual Proof vs. Research Narrative

For recruiters and senior design leaders reviewing portfolios in under 60 seconds, is the balance between upfront metrics and deep architectural breakdown hitting the mark?

👉 Looking for: What draws your eye first in the project sections?